Weather_Graphs

Wacker

2026-04-14

##Some graphs on the data I made while studying for the exam

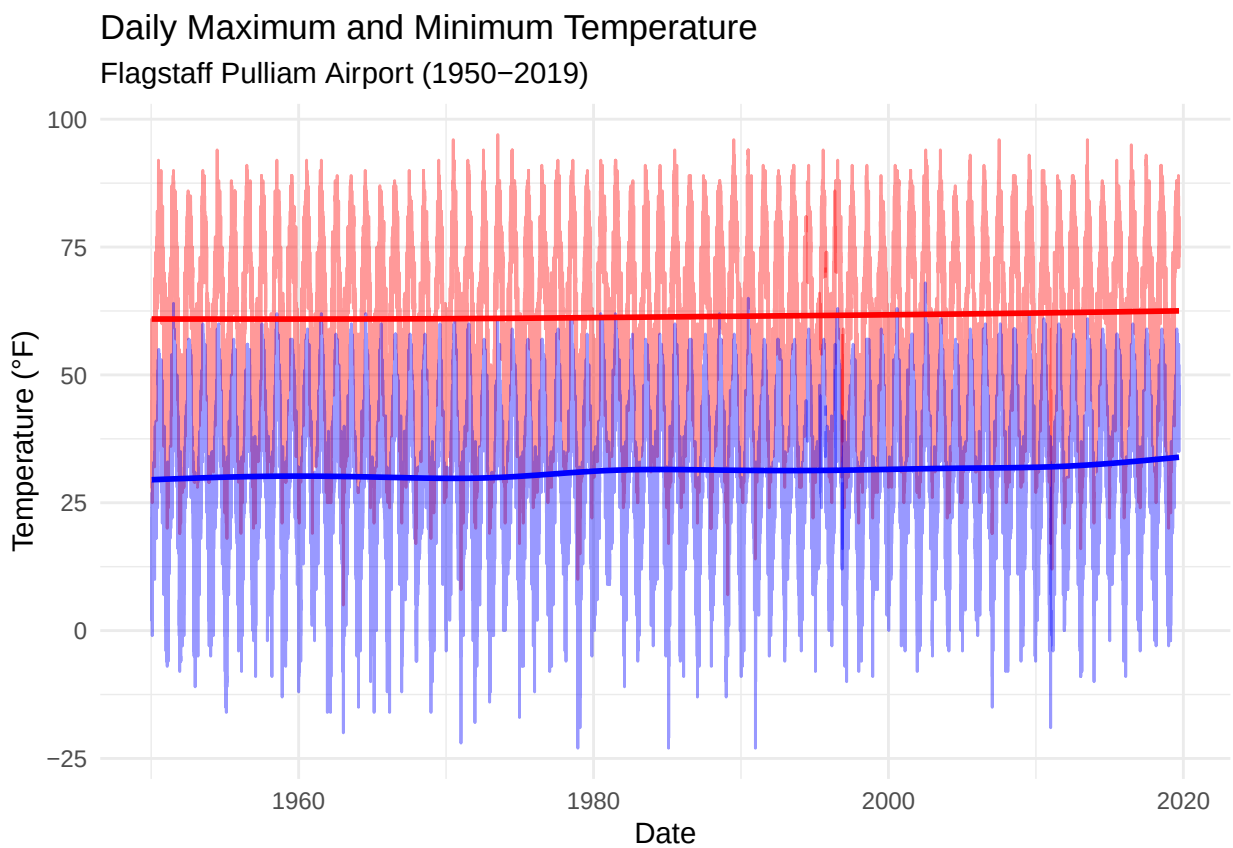
```r
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

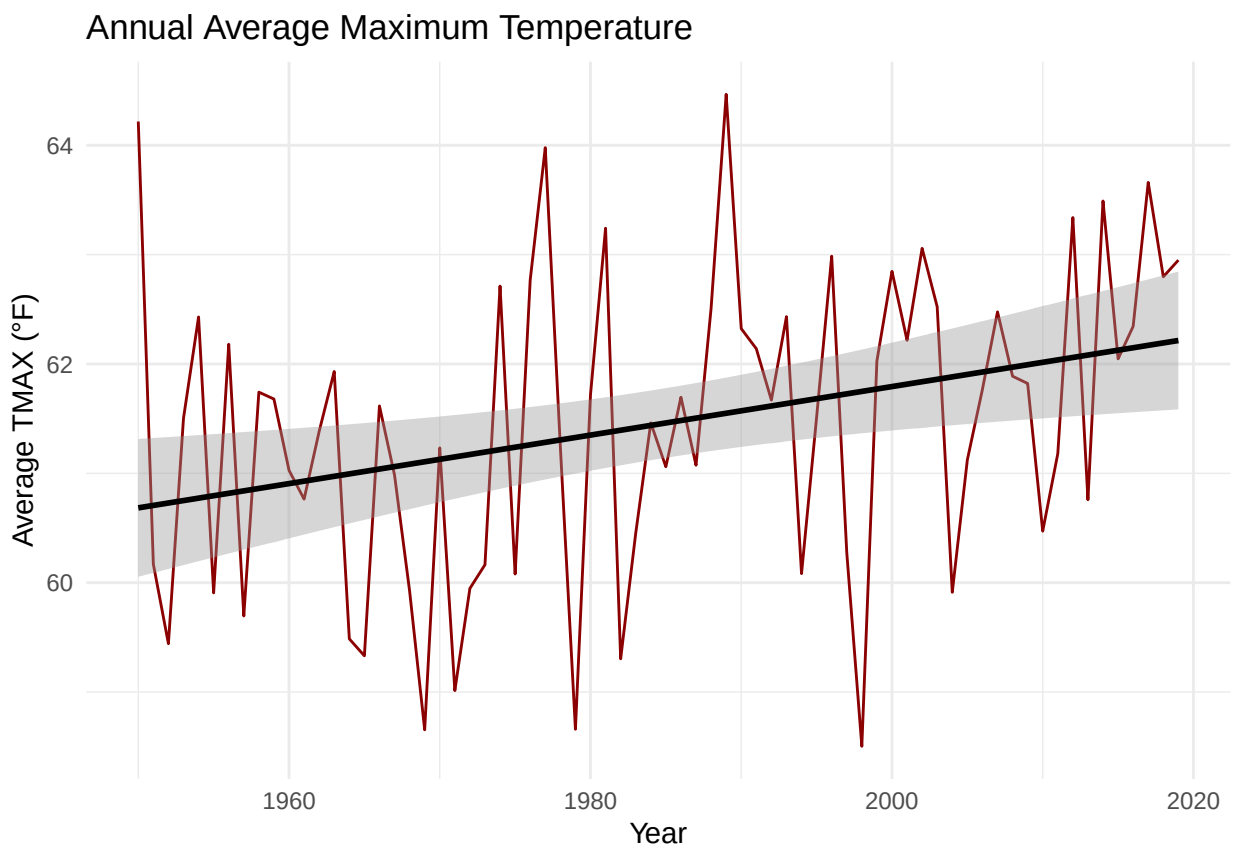
```r
library(ZaileyFlagWeather)
data("Flagstaff_Weather")
```

```r
ggplot(Flagstaff_Weather, aes(x = date)) +
  geom_line(aes(y = tmax), color = "red", alpha = 0.4) +
  geom_line(aes(y = tmin), color = "blue", alpha = 0.4) +
  geom_smooth(aes(y = tmax), color = "red", se = FALSE) +
  geom_smooth(aes(y = tmin), color = "blue", se = FALSE) +
  labs(title = "Daily Maximum and Minimum Temperature",
       subtitle = "Flagstaff Pulliam Airport (1950-2019)",
       x = "Date", y = "Temperature (°F)") +
  theme_minimal()
```



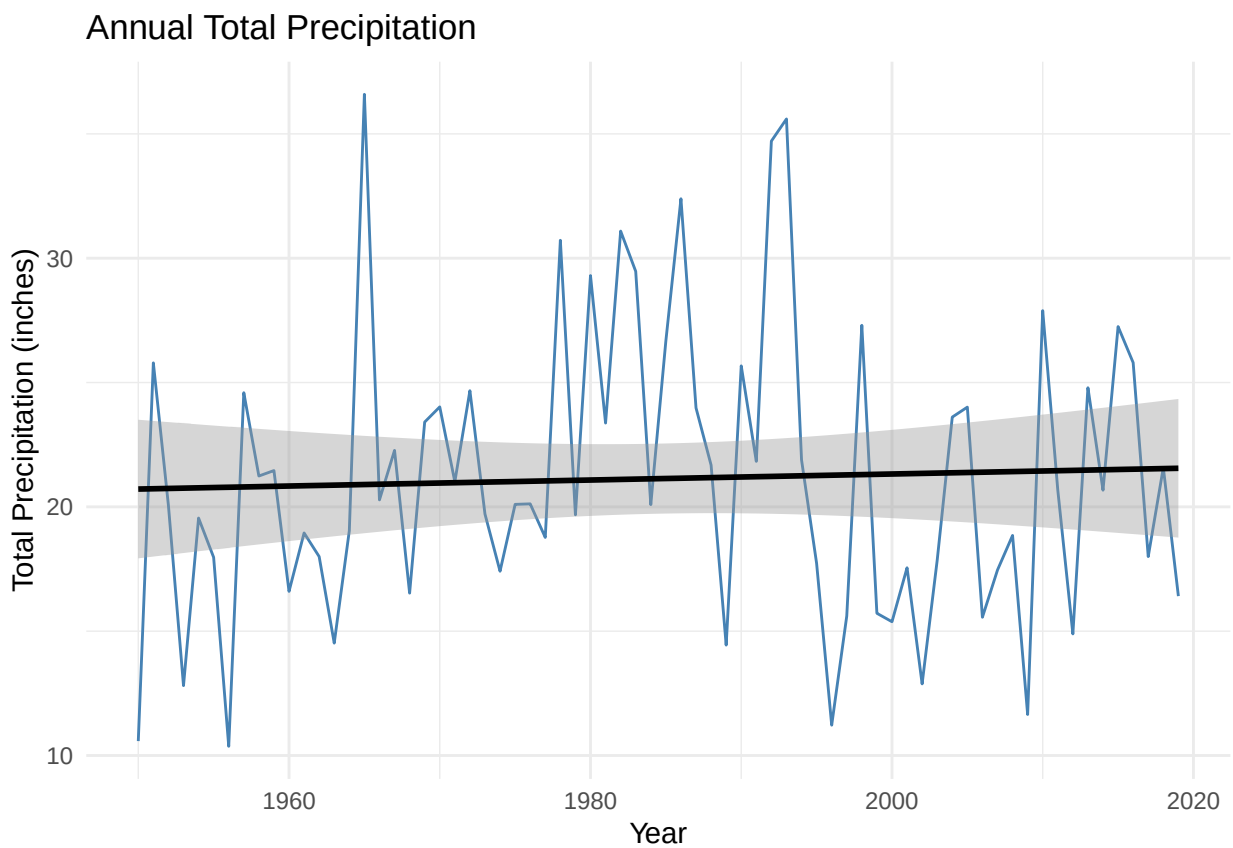

```r
annual_tmax <- Flagstaff_Weather |>
  mutate(year = lubridate::year(date)) |>
  group_by(year) |>
  summarise(avg_tmax = mean(tmax, na.rm = TRUE))

ggplot(annual_tmax, aes(x = year, y = avg_tmax)) +
  geom_line(color = "darkred") +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  labs(title = "Annual Average Maximum Temperature",
       x = "Year",
       y = "Average TMAX (°F)") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

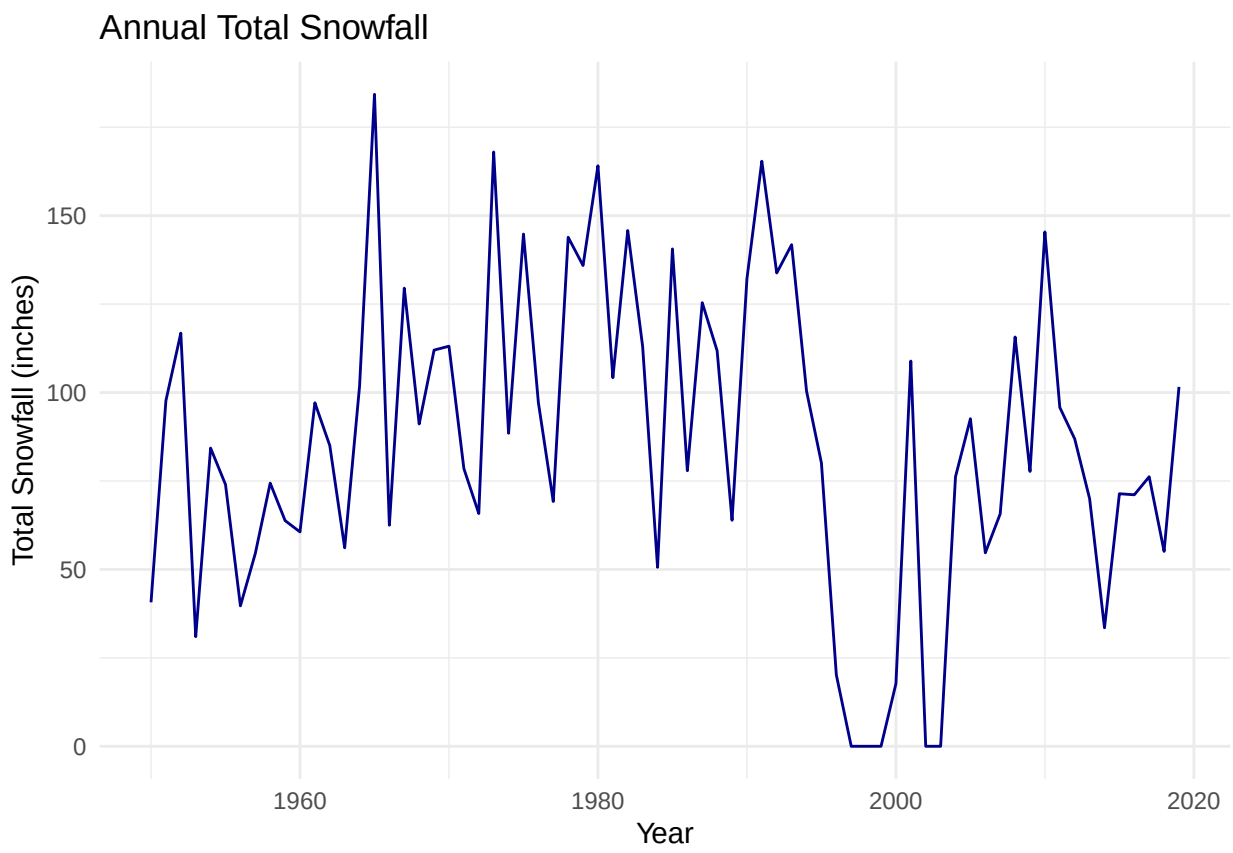



```r
annual_prcp <- Flagstaff_Weather |>
  mutate(year = lubridate::year(date)) |>
  group_by(year) |>
  summarise(total_prcp = sum(prcp, na.rm = TRUE))

ggplot(annual_prcp, aes(x = year, y = total_prcp)) +
  geom_line(color = "steelblue") +
  geom_smooth(method = "lm", color = "black") +
  labs(title = "Annual Total Precipitation",
       x = "Year",
       y = "Total Precipitation (inches)") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
annual_snow <- Flagstaff_Weather |>
  mutate(year = lubridate::year(date)) |>
  group_by(year) |>
  summarise(total_snow = sum(snow, na.rm = TRUE))

ggplot(annual_snow, aes(x = year, y = total_snow)) +
  geom_line(color = "darkblue") +
  labs(title = "Annual Total Snowfall",
       x = "Year",
       y = "Total Snowfall (inches)") +
  theme_minimal()
```


Annual Total Snowfall
Total Snowfall (inches)
150
100
50
0
1960
1980
2000
2020
Year